

ANWESH KABIRAJ

+91 89670 87246 ◇ Kolkata, India

anweshkabiraj@icloud.com ◇ [LinkedIn](#) ◇ [GitHub](#)

GENERATIVE AI ENGINEER | LLM SYSTEMS | RAG | AGENTIC AI

AI Engineer with 3+ years building production-grade RAG pipelines, agentic LLM workflows, and intelligent automation for enterprise and government platforms. Deep expertise across the full GenAI stack — from vector retrieval and reranking to LLMOps, security layers, and scalable inference — with the ability to integrate AI systems end-to-end into modern applications built on React, Next.js, and Node.

EXPERIENCE

AI Engineer, Innovation Labs

Oct 2022 – Present

Tata Consultancy Services

Kolkata, India

- Architected an **enterprise Hybrid RAG pipeline** for Passport Seva policy intelligence using **GPT-5.4 Pro / DeepSeek-V3, NV-Embed-v2, and Qdrant**; implemented **contextual semantic chunking** and **metadata-aware reranking (Qwen3-Reranker-8B)**, cutting hallucinations in long-context policy queries.
- Built an **agentic LLM form-automation system** using **tool calling** and **structured outputs (Zod)**; pipeline: **User Input + RAG → LLM Reasoning → JSON → React Hook Form** — delivering schema-valid, database-ready outputs with near-zero manual intervention.
- Integrated **LLM security and inference layers: Lakera Guard** (prompt injection), **PyTorch FSDP2** (model sharding), and **vLLM PagedAttention** — enabling low-latency, high-throughput production GenAI deployment.
- Built a **custom SpaCy NER pipeline on 120K+ logistics documents**, achieving **89% F1** across 8 entity types; quantized for **edge deployment on x86**, reducing latency **180ms → 10ms** and eliminating central server calls. (*Patent evaluation, TCS 2026*)
- Designed a **lightweight encoder-decoder segmentation model** with a custom **boundary-aware loss**, achieving **91% mIoU**; generalized across 3 domains. (*Patent evaluation, TCS 2026*)
- Engineered a **SARIMA forecasting system** on 10+ years of billing data with **real-time anomaly alerting** (tiered at 10L/50L thresholds) for proactive enterprise-scale financial deviation detection.

SKILLS

GenAI & LLMs	RAG (Hybrid, Agentic), Prompt Engineering, Tool Calling, Structured Outputs, Fine-tuning, LLM Security
Agent Frameworks	LangChain, LlamaIndex, LangGraph
Vector & Retrieval	Qdrant, FAISS, Pinecone, Weaviate, Chroma, Reranking, Hybrid Search
ML & DL	PyTorch, TensorFlow, HuggingFace Transformers, SpaCy, OpenCV, Computer Vision, NER
Inference & LLMOps	vLLM, PyTorch FSDP2, LangSmith, W&B, MLflow, Model Evaluation, A/B Testing
Backend & APIs	FastAPI, Node.js, Express.js, REST, WebSockets
DevOps & Cloud	Docker, Kubernetes, AWS (EC2, S3), Red Hat OpenShift, NGINX
Frontend	React.js, Next.js, Tailwind
Languages	Python, JavaScript, C/C++, Bash

EDUCATION

B.Tech in Information Technology

2018 – 2022

Government College of Engineering and Leather Technology

CGPA: 8.5

RESEARCH & PUBLICATIONS

- CX-Ultraret** — DL model for 13-class thoracic disease detection; **88% accuracy**, 15% above SOTA. [\[Link\]](#)
- CAPCAM Architecture** — Weakly supervised chest X-ray localization; **94% IoBB, 90% Dice score**. [\[Link\]](#)
- GAN-enhanced OCR** — Super-resolution GAN for number plate recognition; accuracy **4–6% → 84–90%**. [\[Link\]](#)